

SmartNPS360 Native API Reference

HTTP APIs used by the Flutter mobile app (Android & iOS)

Base URL: <https://smartnps360.com/>

APIs documented: 9 active endpoints

App version: 1.0.0+66

Last updated: 6 August 2026

Overview

This document lists every backend API endpoint that the SmartNPS360 native app currently calls. Request and response shapes match the live Android and iOS client traffic.

Authenticated endpoints use a Bearer token in the `Authorization` header unless noted otherwise.

ACTIVE ENDPOINTS

1. Sanctum Login
2. Refresh Access Token
3. Duty Heartbeat
4. GPS Ping
5. GPS Batch Upload
6. Register Push Token
7. Delete Push Token
8. Native Permission Status
9. Log Visit Upload

For backend developers: The mobile app does not use `POST /api/gps/point`. That endpoint is not part of the current Android and iOS integration.

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1. Sanctum Login
2. Refresh Access Token
3. Duty Heartbeat
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1. Sanctum Login

POST /api/auth/login

PARAMETERS

Parameter	Type	Required	Description
employee_no	string	Yes	Officer employee number
password	string	Yes	Officer account password
device_name	string	Yes	Client device name (app sends mobile-app)

REQUEST

```
{
  "employee_no": "EMP00123",
  "password": "your-password",
  "device_name": "mobile-app"
}
```

SUCCESS RESPONSE

```
{
  "access_token": "1|exampleAccessToken",
  "refresh_token": "2|exampleRefreshToken",
  "token_type": "Bearer",
  "expires_in": 3600,
  "expires_at": "2026-07-09T12:00:00.000000Z",
  "refresh_expires_in": 2592000,
  "refresh_expires_at": "2026-08-08T12:00:00.000000Z",
  "user": {}
}
```

DESCRIPTION

Authenticates an officer and returns API tokens for native services. Used when the WebView calls `loginWithSanctum` or when iOS needs to mint a bearer token from stored credentials. On success, the app stores access and refresh tokens for later API calls.

The app also accepts `accessToken`, `token`, or `jwt` instead of `access_token`, and `refreshToken` instead of `refresh_token`. The payload may be nested under `data`.

2. Refresh Access Token

POST /api/auth/refresh

PARAMETERS

Parameter	Type	Required	Description
refresh_token	string	Yes	Stored refresh token from login or a previous refresh

REQUEST

```
{
  "refresh_token": "2|exampleRefreshToken"
}
```

SUCCESS RESPONSE

```
{
  "access_token": "3|newAccessToken",
  "token": "3|newAccessToken",
  "token_type": "Bearer",
  "expires_in": 3600,
  "expires_at": "2026-07-09T13:00:00.000000Z",
  "refresh_token": "4|newRefreshToken",
  "refresh_expires_in": 2592000,
  "refresh_expires_at": "2026-08-08T13:00:00.000000Z",
  "user": {}
}
```

DESCRIPTION

Exchanges a valid refresh token for a new access token. Called automatically when the access token is missing or expired, and when an API call returns 401. On success, the app saves the new access token, rotated refresh token, and expiry fields.

3. Duty Heartbeat

GET

/api/heartbeat

PARAMETERS

Parameter	Type	Required	Description
No request body. Requires Bearer authentication.			

REQUEST

```
GET /api/heartbeat
Accept: application/json
```

SUCCESS RESPONSE

```
{
  "status": "on_duty"
}
```

DESCRIPTION

Returns the officer's current duty status. Polled while the app is running to start or stop background GPS tracking. On success, the app reads `on_duty` or `off_duty` from `status`, `duty_status`, `dutyStatus`, `duty`, or `state`, including nested data, payload, or result objects. A plain string response is also supported.

4. GPS Ping

POST /api/gps/ping

PARAMETERS

Parameter	Type	Required	Description
app	string	Yes	App name (SmartNPS360)
platform	string	Yes	android or ios
device_id	string	Yes	Device identifier
deviceId	string	Yes	Same value as device_id
build	string	Yes	App build number
latitude	number	Yes	GPS latitude
longitude	number	Yes	GPS longitude
accuracy	number	Yes	Location accuracy in meters
timestampMs	number	Yes	Fix time in UTC milliseconds
timestamp	string	Yes	Fix time in ISO 8601 UTC
altitude	number	No	Altitude in meters
altitudeAccuracy	number	No	Altitude accuracy
speed	number null	No	Speed in m/s (null when invalid or unavailable)
speedAccuracy	number null	No	Speed accuracy (null when invalid or unavailable)
heading	number null	No	Heading in degrees (null when invalid or unavailable)
headingAccuracy	number null	No	Heading accuracy (null when invalid or unavailable)

Parameter	Type	Required	Description
isMocked	boolean	No	Whether the location is mocked
isSimulatedBySoftware	boolean	No	Whether location is software-simulated
motionActivity	string	No	Fused API label (same engine as Motion Activity screen): stationary, walking, running, cycling, or automotive. Not from speed bands.
motionSource	string	No	Always vehicle_session_fusion
motionFusedState	string	No	Internal fused state (e.g. driving, driving_stopped, walking)
motionNativeActivity	string	No	Raw OS Activity Recognition / Core Motion label
motionNativeConfidence	number	No	OS confidence 0–100
motionSessionActive	boolean	No	Whether sticky vehicle session is active
motionProvisional	boolean	No	True when GPS is filling in before OS catches up
motionReason	string	No	Short fusion reason string
motionSpeedKmh	number	No	Speed used by fusion (smoothed when available)
speedBand	string	No	Speed band label (timing only — not motion)
speedState	string	No	Human-readable speed state
speedKmh	number	No	Raw speed in km/h
smoothedSpeedKmh	number	No	Smoothed speed in km/h
gpsSpeedTrusted	boolean	No	Whether GPS speed is trusted
captureIntervalSeconds	number	No	Capture interval for current speed band
uploadIntervalSeconds	number	No	Upload interval for current speed band
distanceFilterMeters	number	No	Distance filter for current speed band
floor	number	No	Floor level when available

REQUEST

```
{
  "app": "SmartNPS360",
  "platform": "android",
  "device_id": "samsung-galaxy-a12-abc123",
  "deviceId": "samsung-galaxy-a12-abc123",
  "build": "66",
  "latitude": 24.8607,
  "longitude": 67.0011,
  "accuracy": 8.5,
  "altitudeAccuracy": null,
  "timestampMs": 1752494400000,
  "timestamp": "2026-07-14T12:00:00.000Z",
  "altitude": 12.4,
  "speed": 1.2,
  "speedAccuracy": null,
  "heading": null,
  "headingAccuracy": null,
  "isMocked": false,
  "isSimulatedBySoftware": false,
  "motionActivity": "walking",
  "motionSource": "vehicle_session_fusion",
  "motionFusedState": "walking",
  "motionNativeActivity": "walking",
  "motionNativeConfidence": 75,
  "motionSessionActive": false,
  "motionProvisional": false,
  "motionReason": "native walking",
  "motionSpeedKmh": 4.2,
  "speedBand": "2-8 km/h",
  "speedState": "Walking / patrol",
  "speedKmh": 4.5,
  "smoothedSpeedKmh": 4.2,
  "gpsSpeedTrusted": true,
  "captureIntervalSeconds": 10,
  "uploadIntervalSeconds": 10,
  "distanceFilterMeters": 5,
  "floor": null
}
```

SUCCESS RESPONSE

HTTP 2xx

DESCRIPTION

Uploads a single GPS fix immediately while the officer is on duty. Used for real-time location updates. Stationary / below 2 km/h fixes are still uploaded here but are not queued for GPS Batch. On success, the app treats any HTTP 2xx response as accepted.

5. GPS Batch Upload

POST /api/gps/batch

PARAMETERS

Parameter	Type	Required	Description
points	array	Yes	Array of GPS point objects. Each object uses the same fields as GPS Ping (listed below). App sends up to 20 points per request.

POINT OBJECT FIELDS (EACH ITEM IN POINTS)

Parameter	Type	Required	Description
app	string	Yes	App name (SmartNPS360)
platform	string	Yes	android or ios
device_id	string	Yes	Device identifier
deviceId	string	Yes	Same value as device_id
build	string	Yes	App build number
latitude	number	Yes	GPS latitude
longitude	number	Yes	GPS longitude
accuracy	number	Yes	Location accuracy in meters
timestampMs	number	Yes	Fix time in UTC milliseconds
timestamp	string	Yes	Fix time in ISO 8601 UTC
altitude	number	No	Altitude in meters
altitudeAccuracy	number null	No	Altitude accuracy
speed	number null	No	Speed in m/s (null when invalid or unavailable)

Parameter	Type	Required	Description
speedAccuracy	number null	No	Speed accuracy (null when invalid or unavailable)
heading	number null	No	Heading in degrees (null when invalid or unavailable)
headingAccuracy	number null	No	Heading accuracy (null when invalid or unavailable)
isMocked	boolean	No	Whether the location is mocked
isSimulatedBySoftware	boolean null	No	Whether location is software-simulated
motionActivity	string	No	Fused API label (same as GPS Ping / Motion screen): stationary, walking, running, cycling, or automotive. Not from speed bands.
motionSource	string	No	Always vehicle_session_fusion
motionFusedState	string	No	Internal fused state
motionNativeActivity	string	No	Raw OS motion label
motionNativeConfidence	number	No	OS confidence 0–100
motionSessionActive	boolean	No	Whether sticky vehicle session is active
motionProvisional	boolean	No	True when GPS fill-in is provisional
motionReason	string	No	Short fusion reason string
motionSpeedKmh	number	No	Speed used by fusion
speedBand	string	No	Speed band label (timing only — not motion)
speedState	string	No	Human-readable speed state (e.g. Walking / patrol)
speedKmh	number null	No	Raw speed in km/h
smoothedSpeedKmh	number null	No	Smoothed speed in km/h

Parameter	Type	Required	Description
gpsSpeedTrusted	boolean	No	Whether GPS speed is trusted
captureIntervalSeconds	number	No	Capture interval for current speed band
uploadIntervalSeconds	number	No	Upload interval for current speed band
distanceFilterMeters	number	No	Distance filter for current speed band
floor	number null	No	Floor level when available

REQUEST

```
{
  "points": [
    {
      "app": "SmartNPS360",
      "platform": "android",
      "device_id": "samsung-galaxy-a12-abc123",
      "deviceId": "samsung-galaxy-a12-abc123",
      "build": "66",
      "latitude": 24.8607,
      "longitude": 67.0011,
      "accuracy": 8.5,
      "altitudeAccuracy": null,
      "timestampMs": 1752494400000,
      "timestamp": "2026-07-14T12:00:00.000Z",
      "altitude": 12.4,
      "speed": 1.2,
      "speedAccuracy": null,
      "heading": null,
      "headingAccuracy": null,
      "isMocked": false,
      "isSimulatedBySoftware": false,
      "motionActivity": "walking",
      "motionSource": "vehicle_session_fusion",
      "motionFusedState": "walking",
      "motionNativeActivity": "walking",
      "motionNativeConfidence": 75,
      "motionSessionActive": false,
      "motionProvisional": false,
      "motionReason": "native walking",
      "motionSpeedKmh": 4.2,
      "speedBand": "2-8 km/h",
      "speedState": "Walking / patrol",
      "speedKmh": 4.5,
      "smoothedSpeedKmh": 4.2,
    }
  ]
}
```

```
"gpsSpeedTrusted": true,
"captureIntervalSeconds": 10,
"uploadIntervalSeconds": 10,
"distanceFilterMeters": 5,
"floor": null
},
{
  "app": "SmartNPS360",
  "platform": "android",
  "device_id": "samsung-galaxy-a12-abc123",
  "deviceId": "samsung-galaxy-a12-abc123",
  "build": "66",
  "latitude": 24.8610,
  "longitude": 67.0015,
  "accuracy": 7.0,
  "altitudeAccuracy": 3.2,
  "timestampMs": 1752494460000,
  "timestamp": "2026-07-14T12:01:00.000Z",
  "altitude": 12.6,
  "speed": 9.7,
  "speedAccuracy": 0.8,
  "heading": 95.0,
  "headingAccuracy": 5.0,
  "isMocked": false,
  "isSimulatedBySoftware": false,
  "motionActivity": "automotive",
  "motionSource": "vehicle_session_fusion",
  "motionFusedState": "driving",
  "motionNativeActivity": "driving",
  "motionNativeConfidence": 80,
  "motionSessionActive": true,
  "motionProvisional": false,
  "motionReason": "native driving (session)",
  "motionSpeedKmh": 34.8,
  "speedBand": "30-60 km/h",
  "speedState": "Vehicle movement",
  "speedKmh": 35.2,
  "smoothedSpeedKmh": 34.8,
  "gpsSpeedTrusted": true,
  "captureIntervalSeconds": 3,
  "uploadIntervalSeconds": 3,
  "distanceFilterMeters": 3,
  "floor": null
}
]
}
```

SUCCESS RESPONSE

HTTP 2xx

DESCRIPTION

Uploads multiple queued GPS points in one request. Body shape is { "points": [...] }. Each point uses the same fields as GPS Ping. Stationary / below 2 km/h fixes are sent on GPS Ping only and are not included in this queue. The app flushes when at least 2 points are queued and either 1 minute has elapsed or the queue reaches 20 points. On success, the app treats any HTTP 2xx response as accepted and removes the uploaded points from its local queue.

6. Register Push Token

POST /api/push-token

PARAMETERS

Parameter	Type	Required	Description
platform	string	Yes	android or ios
device_id	string	Yes	Device identifier
push_token	string	Yes	FCM push token
app_version	string	Yes	App version string
build	string	Yes	App build number

REQUEST

```
{
  "platform": "android",
  "device_id": "samsung-galaxy-a12-abc123",
  "push_token": "fcm-device-token",
  "app_version": "1.0.0",
  "build": "66"
}
```

SUCCESS RESPONSE

```
HTTP 2xx
```

DESCRIPTION

Registers the device FCM token with the backend for push notifications. Called after login and when the FCM token changes. On iOS, the app may also post through the WebView session when bearer auth is unavailable. On success, any HTTP 2xx response is treated as registered.

7. Delete Push Token

DELETE /api/push-token

PARAMETERS

Parameter	Type	Required	Description
platform	string	Yes	android or ios
device_id	string	Yes	Device identifier
push_token	string	Yes	FCM push token to remove
app_version	string	Yes	App version string
build	string	Yes	App build number

REQUEST

```
{
  "platform": "ios",
  "device_id": "ios-ABCDEF12-3456-7890-ABCD-EF1234567890",
  "push_token": "fcm-device-token",
  "app_version": "1.0.0",
  "build": "66"
}
```

SUCCESS RESPONSE

```
HTTP 2xx
```

DESCRIPTION

Removes a registered FCM token from the backend. Called during logout to stop push delivery to the device. On success, any HTTP 2xx response is treated as deleted.

8. Native Permission Status

POST /api/native-app/permission-status

PARAMETERS

Parameter	Type	Required	Description
platform	string	Yes	android or ios
deviceId	string	Yes	Device identifier
appVersion	string	Yes	App version string
build	string	Yes	App build number
battery_percentage	number null	No	Current battery level (0–100); null if unavailable
low_power_mode	string	No	enabled, disabled, or unknown
permissions	object	Yes	OS permission snapshot (see below)
checkedAt	string	Yes	Snapshot time in ISO 8601 UTC
app_cycle	string	No	App lifecycle state (e.g. resumed, paused, inactive, detached, hidden)

PERMISSIONS OBJECT

Field	Values	Description
foregroundLocation	granted, denied, unknown	Foreground location permission
backgroundLocation	granted, denied, unknown	Background location permission

Field	Values	Description
preciseLocation	granted, denied, unknown	Precise location permission
notifications	granted, denied, unknown	Notification permission
motionActivity	granted, denied, unknown	Android Activity Recognition / iOS Motion & Fitness permission
batteryOptimization	granted, unknown	Android battery optimization exemption (granted = exempt; unknown = default optimized state or iOS)
backgroundAppRefresh	enabled, disabled, restricted, unknown	iOS Background App Refresh (UIBackgroundRefreshStatus). Android closest equivalent: app background restriction (enabled = not restricted; disabled = restricted; restricted is iOS-only)
push	enabled, disabled	In-app push toggle state (when uploaded)

REQUEST

```
{
  "platform": "android",
  "deviceId": "samsung-galaxy-a12-abc123",
  "appVersion": "1.0.0",
  "build": "66",
  "battery_percentage": 78,
  "low_power_mode": "disabled",
  "permissions": {
    "foregroundLocation": "granted",
    "backgroundLocation": "granted",
    "preciseLocation": "granted",
    "notifications": "granted",
    "motionActivity": "granted",
    "batteryOptimization": "unknown",
    "backgroundAppRefresh": "enabled",
    "push": "enabled"
  },
  "checkedAt": "2026-07-14T12:00:00.000Z",
}
```

```
"app_cycle": "resumed"  
}
```

SUCCESS RESPONSE

HTTP 2xx

DESCRIPTION

Reports the device's current OS permission and battery snapshot to the backend. Sent when permissions change, when battery or low-power mode changes, when the app lifecycle changes, or when the in-app push toggle changes. On success, any HTTP 2xx response is treated as accepted.

9. Log Visit Upload

POST /api/visits

One multipart request only — meta JSON plus media (and optional voice) files together. Not separate calls. Not a pure JSON body.

FORM PARTS

Part	Type	Required	Description
meta	JSON string	Yes	Visit metadata, attention-needed flag, and per-item notes/GPS
media[0], media[1], ...	file (photo/video)	Yes (at least one)	Media files indexed by <code>client_index</code>
voice[0], voice[2], ...	file (audio)	Only if item has a voice note	Voice note for the matching <code>client_index</code>
attention_voice	file (audio)	Only if attention needed has a voice note	Visit-level Attention Needed voice note (not tied to a media item)

`media[N]` / `voice[N]` must match `items[].client_index`. `attention_voice` is independent of item indexes.

META JSON FIELDS

Parameter	Type	Required	Description
<code>client_draft_id</code>	string (UUID)	Yes	Idempotency key for this report; same value on retry after kill/reopen until success or discard
<code>started_at</code>	string	Yes	Visit start time (ISO 8601 UTC)
<code>submitted_at</code>	string	Yes	Upload submit time (ISO 8601 UTC)
<code>region_id</code>	number	Yes	Patrol region id (source of truth)
<code>site_id</code>	number	Yes	Patrol site id (source of truth)

Parameter	Type	Required	Description
region_name	string	No	Display-only region name
site_name	string	No	Display-only site name
schedule_id	number	No	Schedule id when the visit was opened from a schedule
attention_needed	object	Yes	Always present. Visit-level Attention Needed toggle + optional text/voice note
items	array	Yes	Per-media metadata in draft order (client_index 0..n-1)

ATTENTION_NEEDED OBJECT FIELDS

Parameter	Type	Required	Description
needed	string	Yes	"yes" or "no" from the Attention Needed toggle
text_note	string	Yes	Attention Needed text note when needed is "yes"; otherwise empty string
has_voice_note	boolean	Yes	If true (and needed is "yes"), attention_voice must be present in the same request

ITEMS[] OBJECT FIELDS

Parameter	Type	Required	Description
client_index	number	Yes	0-based index; must match media [N] / optional voice [N]
type	string	Yes	photo or video
text_note	string	No	Officer text note for this media item (may be empty)
captured_at	string	No	Capture time (ISO 8601 UTC)
latitude	number	No	Capture GPS latitude
longitude	number	No	Capture GPS longitude
accuracy_meters	number	No	GPS accuracy in meters

Parameter	Type	Required	Description
has_voice_note	boolean	Yes	If true, voice [N] must be present in the same request

REQUEST (MULTIPART/FORM-DATA)

```

POST /api/visits
Authorization: Bearer <access_token>
Content-Type: multipart/form-data
Accept: application/json

meta                → JSON string (see below)
media[0]            → media_0.jpg          (image/jpeg)
media[1]            → media_1.mp4          (video/mp4)
media[2]            → media_2.jpg          (image/jpeg)
voice[0]            → voice_0.m4a          (audio/mp4)
voice[2]            → voice_2.m4a          (audio/mp4)
attention_voice     → attention_voice.m4a  (audio/mp4) ← only when
attention_needed.has_voice_note is true

```

META EXAMPLE

```

{
  "client_draft_id": "8f3c2a1e-6b44-4d91-9c2e-1a7b0f5d3e21",
  "started_at": "2026-08-02T11:10:11.234Z",
  "submitted_at": "2026-08-02T11:35:02.100Z",
  "region_id": 12,
  "site_id": 845,
  "region_name": "Lahore North",
  "site_name": "Gate A",
  "schedule_id": 4012,
  "attention_needed": {
    "needed": "yes",
    "text_note": "Gate lock was forced open",
    "has_voice_note": false
  },
  "items": [
    {
      "client_index": 0,
      "type": "photo",
      "text_note": "Main gate lock was open",
      "captured_at": "2026-08-02T11:12:01.000Z",
      "latitude": 31.520416,
      "longitude": 74.358748,
      "accuracy_meters": 8.5,
      "has_voice_note": true
    },
  ],
}

```



```

{
  "client_index": 1,
  "type": "video",
  "text_note": "",
  "captured_at": "2026-08-02T11:18:44.000Z",
  "latitude": 31.520901,
  "longitude": 74.359112,
  "accuracy_meters": 12.0,
  "has_voice_note": false
},
{
  "client_index": 2,
  "type": "photo",
  "text_note": "Side fence damage",
  "captured_at": "2026-08-02T11:24:10.500Z",
  "latitude": 31.521055,
  "longitude": 74.359480,
  "accuracy_meters": 6.2,
  "has_voice_note": true
}
]
}

```

ATTENTION_NEEDED VARIANTS

```

// Toggle ON + text note (no attention_voice file)
"attention_needed": {
  "needed": "yes",
  "text_note": "Gate lock was forced open",
  "has_voice_note": false
}

// Toggle ON + voice note (send attention_voice file)
"attention_needed": {
  "needed": "yes",
  "text_note": "",
  "has_voice_note": true
}

// Toggle OFF (always still sent; no attention_voice file)
"attention_needed": {
  "needed": "no",
  "text_note": "",
  "has_voice_note": false
}

```

SUCCESS RESPONSE

```
{
  "success": true,
  "visit_id": 10921,
  "client_draft_id": "8f3c2a1e-6b44-4d91-9c2e-1a7b0f5d3e21",
  "items_saved": 3,
  "message": "Visit report uploaded successfully"
}
```

ERROR RESPONSE

```
{
  "success": false,
  "message": "Validation failed",
  "errors": {
    "site_id": ["The selected site is invalid."],
    "items.0.voice": ["Voice note missing for client_index 0."],
    "attention_voice": ["Attention needed voice note missing."]
  }
}
```

RULES

- `client_draft_id` is a UUID used for idempotency — the same id must not create a duplicate visit.
- `region_id` and `site_id` are required and are the source of truth; names are display-only.
- `client_index` is `0..n-1` in the same order as the saved draft list.
- `type` is `photo` or `video`.
- Timestamps are ISO 8601 UTC.
- `attention_needed` is always present in meta. needed is "yes" or "no".
- When `attention_needed.needed` is "no", `text_note` is "" and `has_voice_note` is false.
- When `attention_needed.needed` is "yes" and `has_voice_note` is true, the multipart part `attention_voice` must be present.
- If an item's `has_voice_note` is true, `voice[N]` must exist for that index. No `voice[1]` in the example because item 1 has `has_voice_note: false`.
- Typical media content types sent by the app: `image/jpeg`, `image/png`, `image/webp`, `image/heic`, `video/mp4`, `video/quicktime`; voice notes usually `audio/mp4` (.m4a), also

audio/mpeg, audio/wav, audio/aac.

- No client-enforced max file size / count limit beyond requiring at least one media item.

DESCRIPTION

Uploads a patrol visit report from the Log Visit flow as a single multipart request. Requires Bearer authentication (same Sanctum access token as other native APIs; 401 triggers the shared refresh-and-retry path). Includes an always-present `attention_needed` object for the visit-level Attention Needed toggle (optional text or voice). On success, the app reads `success`, `visit_id`, `client_draft_id`, `items_saved`, and `message` (camelCase aliases are also accepted).